

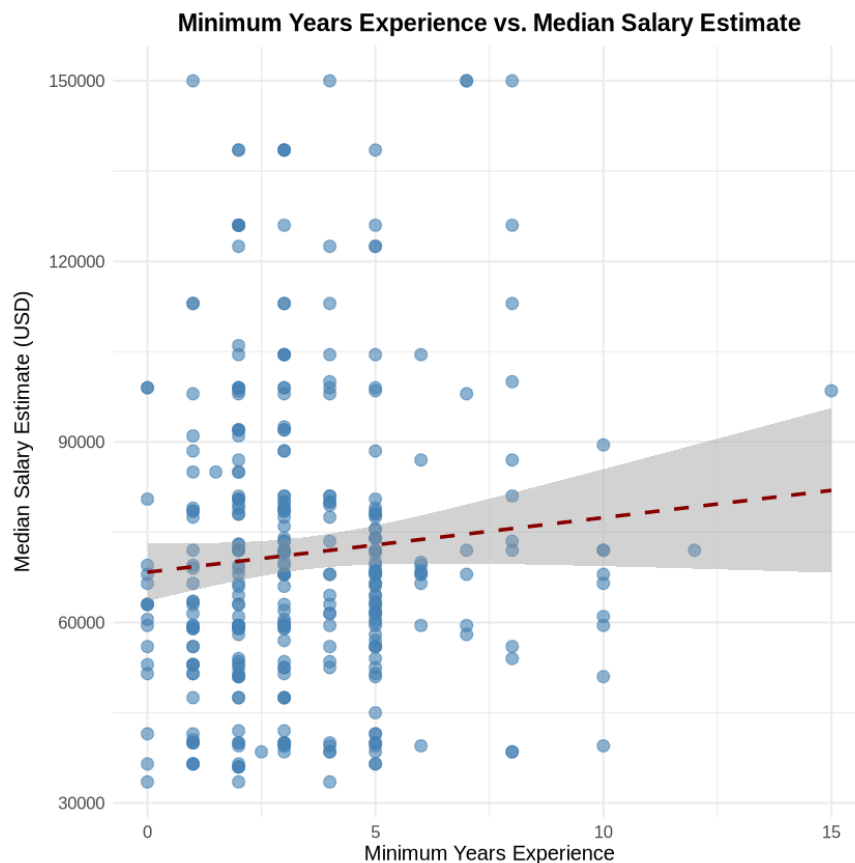
Data Analyst Job Insights - Using OpenRouter AI & Google Sheets

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This report shows preliminary job market analysis for data analysis roles. This project involved using [OpenRouter.AI](#) and Google Sheets to efficiently organize, clean, and analyze a large subset of job market data from Glassdoor. The full dataset can be found here: [Original Data Source](#). Finally the [Julius.AI](#) was used to create plots and identify role requirement trends pertaining to (1) *Data Language* and (2) *Years of Experience*.

Minimum Years of Experience vs. Salary:

The scatter plot below shows the relationship between years of experience required and salary. There's a positive correlation - as minimum years of experience increases, median salary tends to increase as well. The dashed red line shows the linear trend. However, there's considerable variation, with salaries ranging quite widely at each experience level. Most positions require between 0-10 years of experience, with salaries ranging from about \$40,000 to \$150,000 USD.

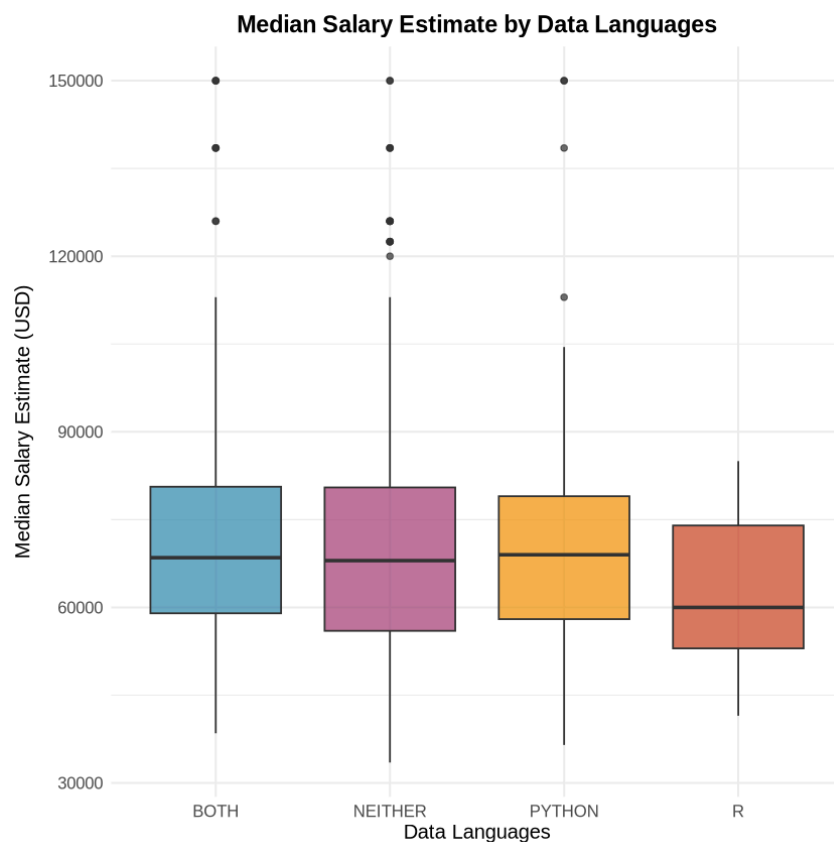


Data Language Need vs. Salary:

The box plot below compares salaries across different data language requirements:

- **BOTH (Python & R):** Shows the highest median salary and widest range, suggesting positions requiring both languages command premium compensation
- **PYTHON:** Has a relatively high median salary with moderate variation
- **R:** Shows a similar median to Python but with a smaller sample size (only 12 positions)
- **NEITHER:** Has the lowest median salary and tightest distribution, indicating positions not requiring programming languages tend to pay less

There's a clear salary premium for positions requiring data programming skills, especially when both Python and R are required. The "NEITHER" category has notably lower salaries, highlighting the value of technical programming skills in data analyst roles.



Reflection:

This project was super helpful and makes me optimistic for future efforts where I have to organize, sanitize, and/or analyze large sets of data. Before starting a project like this, it is first important to know which AI model to use based on the context of the project so factors like data sensitivity / sharing limitations should be thoughtfully considered. Other than this, using OpenRouter AI, GoogleSheets and AppsScript were super handy and easy to use.